

I. Pen-and-paper

Pen-and-paper

$d=2$, vetores bivariados

$$\textcircled{1} \quad n_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, n_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, n_3 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$\mu_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \mu_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \Sigma_1 = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}, \Sigma_2 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \pi_1 = 0.5, \pi_2 = 0.5$$

First epoch:

E-step:

$$n_1: P(K|n_1) = \frac{P(n_1|K)P(K)}{P(n_1)}$$

$K=1$:

$$P(n_1|K=1) \sim N(\mu_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \Sigma_1 = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}) = \frac{1}{2\pi\sqrt{|\Sigma_1|}} e^{-\frac{1}{2}(n_1 - \mu_1)^T \Sigma_1^{-1} (n_1 - \mu_1)}$$

$$\cdot |\Sigma_1| = \det \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} = 4 \times 4 - 1 \times 1 = 16 - 1 = 15$$

$$\cdot \Sigma_1^{-1} = \frac{1}{\det} \times \begin{bmatrix} 4 & -1 \\ -1 & 4 \end{bmatrix} = \frac{1}{15} \begin{bmatrix} 4 & -1 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} 0.2667 & -0.0667 \\ -0.0667 & 0.2667 \end{bmatrix}$$

$$\cdot n_1 - \mu_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$\frac{1}{2\pi\sqrt{15}} e^{-\frac{1}{2} \begin{bmatrix} 0 \\ -1 \end{bmatrix}^T \begin{bmatrix} 0.2667 & -0.0667 \\ -0.0667 & 0.2667 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \end{bmatrix}} = \frac{1}{2\pi\sqrt{15}} e^{-\frac{1}{2} \begin{bmatrix} 0 & -1 \end{bmatrix} \cdot \begin{bmatrix} -0.0667 \\ 0.2667 \end{bmatrix}} = \frac{1}{2\pi\sqrt{15}} e^{-\frac{1}{2} \cdot \frac{1}{3}} = \frac{1}{2\pi\sqrt{15}} e^{-\frac{1}{6}} \approx 0.02945$$

$$\cdot \text{Posterior}_1 = P(n_1|K=1) \times \text{Prior}_1 = 0.02945 \times 0.5 = 0.014725$$

$K=2$:

$$P(n_1|K=2) \sim N(\mu_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \Sigma_2 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}) = \frac{1}{2\pi\sqrt{|\Sigma_2|}} e^{-\frac{1}{2}(n_1 - \mu_2)^T \Sigma_2^{-1} (n_1 - \mu_2)}$$

$$\cdot |\Sigma_2| = \det \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 2 \times 2 - 0 = 4$$

$$\cdot \Sigma_2^{-1} = \frac{1}{4} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix}$$

$$\cdot n_1 - \mu_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$\frac{1}{2\pi\sqrt{4}} e^{-\frac{1}{2} \begin{bmatrix} 0 & -1 \end{bmatrix}^T \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \end{bmatrix}} = \frac{1}{4\pi} e^{-\frac{1}{2} \begin{bmatrix} 0 & -1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ -1/2 \end{bmatrix}} = \frac{1}{4\pi} e^{-\frac{1}{2} \cdot \frac{1}{2}} = \frac{1}{4\pi} e^{-\frac{1}{4}} \approx 0.061975$$

$$\cdot \text{Posterior}_2 = P(n_1|K=2) \times \text{Prior}_2 = 0.061975 \times 0.5 = 0.0309875$$

$$\cdot P(K=1|n_1) = \frac{0.014725}{0.014725 + 0.0309875} = 0.322084, \quad \cdot P(K=2|n_1) = \frac{0.0309875}{0.014725 + 0.0309875} = 0.677916$$

① cont.

$$\underline{n_2}: p(k | n_2) = \frac{p(n_2 | k) p(k)}{p(n_2)}$$

• $K=1$:

$$p(n_2 | K=1) \sim N(\mu_1 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \Sigma_1 = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}) = \frac{1}{2\pi\sqrt{|\Sigma_1|}} \cdot e^{-1/2 (n_2 - \mu_1)^T \cdot \Sigma_1^{-1} (n_2 - \mu_1)}$$

$$\bullet |\Sigma_1| = \det \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} = 15$$

$$\bullet \Sigma_1^{-1} = \frac{1}{15} \begin{bmatrix} 4 & -1 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} \frac{4}{15} & -\frac{1}{15} \\ -\frac{1}{15} & \frac{4}{15} \end{bmatrix}$$

$$\bullet n_2 - \mu_1 = \begin{bmatrix} 0 \\ 2 \end{bmatrix} - \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$$

$$= \frac{1}{2\pi\sqrt{15}} \cdot e^{-1/2 \begin{bmatrix} -2 & 3 \end{bmatrix} \begin{bmatrix} \frac{4}{15} & -\frac{1}{15} \\ -\frac{1}{15} & \frac{4}{15} \end{bmatrix} \cdot \begin{bmatrix} -2 \\ 3 \end{bmatrix}} = \frac{1}{2\pi\sqrt{15}} \cdot e^{-1/2 \begin{bmatrix} 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} -\frac{11}{15} \\ \frac{14}{15} \end{bmatrix}} =$$

$$= \frac{1}{2\pi\sqrt{15}} \cdot e^{-1/2 \cdot \left(\frac{64}{15}\right)} = \frac{e^{-32/15}}{2\pi\sqrt{15}} \approx 0,0048674$$

$$\bullet \text{Posterior}_1 = p(n_2 | K=1) \times \text{Prior}_1 = 0,0048674 \times 0,5 = 0,0024337$$

• $K=2$:

$$p(n_2 | K=2) \sim N(\mu_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \Sigma_2 = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}) = \frac{1}{2\pi\sqrt{|\Sigma_2|}} \cdot e^{-1/2 (n_2 - \mu_2)^T \cdot \Sigma_2^{-1} (n_2 - \mu_2)}$$

$$\bullet |\Sigma_2| = \det \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} = 2$$

$$\bullet \Sigma_2^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\bullet n_2 - \mu_2 = \begin{bmatrix} 0 \\ 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{2\pi\sqrt{2}} \cdot e^{-1/2 \begin{bmatrix} -1 & 1 \end{bmatrix} \begin{bmatrix} 1/2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix}} = \frac{1}{4\pi} \cdot e^{-1/2 \begin{bmatrix} -1 & 1 \end{bmatrix} \begin{bmatrix} -1/2 \\ 1 \end{bmatrix}} = \frac{1}{4\pi} \cdot e^{-1/2 \cdot 1} = \frac{1}{4\pi} \cdot e^{-1/2} =$$

$$\approx 0,0482664$$

$$\text{Posterior}_2 = p(n_2 | K=2) \times \text{Prior}_2 = 0,0482664 \times 0,5 = 0,0241332$$

$$\bullet \frac{p(K=1 | n_2)}{0,0024337 + 0,0241332} = \frac{0,0024337}{0,0265669} = 0,091604$$

$$\bullet \frac{p(K=2 | n_2)}{0,0024337 + 0,0241332} = \frac{0,0241332}{0,0265669} = 0,908396$$

① cont.

$$\frac{n_3}{K=1} \cdot p(K|n_3) = \frac{p(n_3|K) \cdot p(K)}{p(n_3)}$$

$$p(n_3|K=1) \sim N(\mu_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \Sigma_1 = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}) = \frac{1}{2\pi\sqrt{|\Sigma_1|}} \cdot e^{-1/2 \cdot (n_3 - \mu_1)^T \cdot \Sigma_1^{-1} \cdot (n_3 - \mu_1)}$$

$$\cdot |\Sigma_1| = \det \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} = 15$$

$$\cdot \Sigma_1^{-1} = \frac{1}{\det} \cdot \begin{bmatrix} 4 & -1 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} \frac{4}{15} & -\frac{1}{15} \\ -\frac{1}{15} & \frac{4}{15} \end{bmatrix}$$

$$\cdot n_3 - \mu_1 = \begin{bmatrix} 3 \\ -1 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\frac{1}{2\pi\sqrt{15}} \cdot e^{-1/2 \cdot \begin{bmatrix} 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} \frac{4}{15} & -\frac{1}{15} \\ -\frac{1}{15} & \frac{4}{15} \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix}} = \frac{1}{2\pi\sqrt{15}} \cdot e^{-1/2 \cdot \begin{bmatrix} 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} \frac{4}{15} \\ -\frac{1}{15} \end{bmatrix}} = \frac{1}{2\pi\sqrt{15}} \cdot e^{-1/2 \cdot \frac{4}{15}}$$

$$= \frac{1}{2\pi\sqrt{15}} \cdot e^{-\frac{2}{15}} \approx 0,035964$$

$$\cdot \text{Posterior}_1 = p(n_3|K=1) \times \text{Prior}_1 = 0,035964 \times 0,5 = 0,017982$$

K=2

$$p(n_3|K=2) \sim N(\mu_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \Sigma_2 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}) = \frac{1}{2\pi\sqrt{|\Sigma_2|}} \cdot e^{-1/2 \cdot (n_3 - \mu_2)^T \cdot \Sigma_2^{-1} \cdot (n_3 - \mu_2)}$$

$$\cdot |\Sigma_2| = \det \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 4$$

$$\cdot \Sigma_2^{-1} = \frac{1}{4} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix}$$

$$\cdot n_3 - \mu_2 = \begin{bmatrix} 3 \\ -1 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

$$\frac{1}{2\pi\sqrt{4}} \cdot e^{-1/2 \cdot \begin{bmatrix} 2 & -2 \end{bmatrix} \cdot \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ -2 \end{bmatrix}} = \frac{1}{4\pi} \cdot e^{-1/2 \cdot \begin{bmatrix} 2 & -2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix}} = \frac{1}{4\pi} \cdot e^{-1/2 \cdot 4} = \frac{1}{4\pi} \cdot e^{-2}$$

$$\approx 0,01077$$

$$\cdot \text{Posterior}_2 = p(n_3|K=2) \times \text{Prior}_2 = 0,01077 \times 0,5 = 0,005385$$

$$\cdot p(K=1|n_3) = \frac{0,017982}{0,017982 + 0,005385} = 0,76953$$

$$\cdot p(K=2|n_3) = \frac{0,005385}{0,005385 + 0,017982} = 0,23047$$

(1) cont.

Van-der-Waerden

$$\mu_L = \frac{\sum_{i=1}^N P(L|n_i) \cdot n_i}{\sum_{i=1}^N P(L|n_i)}$$

$$\sum_L^{ij} = \frac{\sum_{k=1}^N P(L|n_k) \cdot (n_k^i - \mu_L^i) (n_k^j - \mu_L^j)}{\sum_{k=1}^N P(L|n_k)}$$

$$\text{Prior}_L = P(L=k) = \frac{\sum_{j=1}^K P(L=j|n_i)}{N}$$

M-Step

L=1

$$N_1 = \sum_{i=1}^N P(L|n_i) = 0,322084 + 0,091604 + 0,769553 = 1,18324$$

$$\mu_1 = \frac{0,322084 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0,091604 \cdot \begin{bmatrix} 0 \\ 2 \end{bmatrix} + 0,769553 \cdot \begin{bmatrix} 3 \\ -1 \end{bmatrix}}{1,18324} = \begin{bmatrix} 2,22334 \\ -0,495543 \end{bmatrix}$$

$$\sum^{(1,1)} = \frac{0,322084 (1-2,22334)^2 + 0,091604 (0-2,22334)^2 + 0,769553 (3-2,22334)^2}{1,18324} \approx 1,18237$$

$$\sum^{(2,1)} = \sum^{(1,2)} = \frac{0,322084 (1-2,22334)(0+0,495543) + 0,091604 (0-2,22334)(2+0,495543) + 0,769553 (3-2,22334)(-1+0,495543)}{1,18324} =$$

$$\approx -0,849374$$

$$\sum^{(2,2)} = \frac{0,322084 (0+0,495543)^2 + 0,091604 (2+0,495543)^2 + 0,769553 (-1+0,495543)^2}{1,18324} =$$

$$\approx 0,714485$$

Portanto, $\sum_1^{\text{atualizado}} = \begin{bmatrix} 1,18237 & -0,849374 \\ -0,849374 & 0,714485 \end{bmatrix}$, e $\pi_1 = \frac{1,18324}{2} = 0,394413$

L=2

$$N_2 = \sum_{i=1}^N P(L|n_i) = 0,677916 + 0,908396 + 0,230447 = 1,81676$$

$$\mu_2 = \frac{0,677916 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0,908396 \cdot \begin{bmatrix} 0 \\ 2 \end{bmatrix} + 0,230447 \cdot \begin{bmatrix} 3 \\ -1 \end{bmatrix}}{1,81676} = \begin{bmatrix} 0,753681 \\ 0,873173 \end{bmatrix}$$

① cont.

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$$\sum^{(1,1)} = \frac{0,677916(1-0,753681)^2 + 0,908396(0-0,753681)^2 + 0,230447(3-0,753681)^2}{1,87676}$$

$$\approx 0,946716$$

$$\sum^{(2,1)} = \sum^{(1,2)} = \frac{0,677916(1-0,753681)(0-0,873173) + 0,908396(0-0,753681)(2-0,873173) + 0,230447(3-0,753681)(-1-0,873173)}{1,87676}$$

$$\approx -1,038634$$

$$\sum^{(2,2)} = \frac{0,677916(0-0,873173)^2 + 0,908396(2-0,873173)^2 + 0,230447(-1-0,873173)^2}{1,87676}$$

$$\approx 1,36445$$

Portonato, $\sum_2^{\text{actualizado}} = \begin{bmatrix} 0,946716 & -1,038634 \\ -1,038634 & 1,36445 \end{bmatrix}$

$$\bar{\pi}_2 = \frac{1,87676}{2} = 0,605587$$

To summarize, the first epoch updated values are:

$$N_1 = 1,1824$$

$$N_2 = 1,87676$$

$$\mu_1 = \begin{bmatrix} 2,233 \\ -0,496 \end{bmatrix}$$

$$\mu_2 = \begin{bmatrix} 0,753 \\ 0,873 \end{bmatrix}$$

$$\Sigma_1 = \begin{bmatrix} 1,1824 & -0,8494 \\ -0,8494 & 0,7145 \end{bmatrix}$$

$$\Sigma_2 = \begin{bmatrix} 0,9467 & -1,0386 \\ -1,0386 & 1,3645 \end{bmatrix}$$

$$\pi_1 = 0,3944$$

$$\bar{\pi}_2 = 0,60559$$

Second epoch:

E-step:

$$\underline{n}_1: p(k|n_1) = \frac{p(n_1|k)p(k)}{p(n_1)}$$

$$k=1$$

$$p(n_1|k=1) = N\left(\mu_1 = \begin{bmatrix} 2,223 \\ -0,496 \end{bmatrix}, \Sigma_1 = \begin{bmatrix} 1,182 & -0,849 \\ -0,849 & 0,714 \end{bmatrix}\right) = \frac{1}{2\pi\sqrt{|\Sigma_1|}} \cdot e^{-\frac{1}{2}(n_1 - \mu_1)^T \cdot \Sigma_1^{-1} \cdot (n_1 - \mu_1)}$$

① cont.

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$$|\Sigma_1| = \det \begin{bmatrix} 1,182 & -0,849 \\ 0,849 & 0,714 \end{bmatrix} = 0,123$$

$$\Sigma_1^{-1} = \frac{1}{0,123} \begin{bmatrix} 0,714 & 0,849 \\ 0,849 & 1,182 \end{bmatrix} = \begin{bmatrix} 5,805 & 6,886 \\ 6,886 & 9,618 \end{bmatrix}$$

$$n_1 - \mu_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 2,223 \\ -0,496 \end{bmatrix} = \begin{bmatrix} -1,223 \\ 0,496 \end{bmatrix}$$

$$\frac{1}{2\pi\sqrt{0,123}} \cdot e^{-1/2 \begin{bmatrix} -1,223 & 0,496 \end{bmatrix} \cdot \begin{bmatrix} 5,805 & 6,886 \\ 6,886 & 9,618 \end{bmatrix} \cdot \begin{bmatrix} -1,223 \\ 0,496 \end{bmatrix}} \approx \frac{1}{2\pi\sqrt{0,123}} \cdot e^{-1/2 \cdot 7,419} \approx$$

$$\approx 0,011$$

$$\text{Posterior}_1 = p(n_1 | k=1) \times \text{Prior}_1 = 0,011 \times 0,394 = 0,00438$$

$$K=2: p(n_1 | k=2) \sim N \left(\mu_2 = \begin{bmatrix} 0,754 \\ 0,873 \end{bmatrix}, \Sigma_2 = \begin{bmatrix} 0,947 & -1,0387 \\ -1,0387 & 1,364 \end{bmatrix} \right) = \frac{1}{2\pi\sqrt{|\Sigma_2|}} \cdot e^{-1/2 (n_1 - \mu_2)^T \Sigma_2^{-1} (n_1 - \mu_2)}$$

$$|\Sigma_2| = \det \begin{bmatrix} 0,947 & -1,0387 \\ -1,0387 & 1,364 \end{bmatrix} = 0,213$$

$$\Sigma_2^{-1} = \frac{1}{0,213} \begin{bmatrix} 1,364 & 1,0387 \\ 1,0387 & 0,947 \end{bmatrix} = \begin{bmatrix} 6,406 & 4,876 \\ 4,876 & 4,448 \end{bmatrix}$$

$$n_1 - \mu_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0,754 \\ 0,873 \end{bmatrix} = \begin{bmatrix} 0,246 \\ -0,873 \end{bmatrix}$$

$$\frac{1}{2\pi\sqrt{0,213}} \cdot e^{-1/2 \cdot \begin{bmatrix} 0,246 & -0,873 \end{bmatrix} \cdot \begin{bmatrix} 6,406 & 4,876 \\ 4,876 & 4,448 \end{bmatrix} \cdot \begin{bmatrix} 0,246 \\ -0,873 \end{bmatrix}} = \frac{1}{2\pi\sqrt{0,213}} \cdot e^{-1/2 \cdot 1,6799} = 0,14888$$

$$\text{Posterior}_2 = p(n_1 | k=2) \times \text{Prior}_2 = 0,14888 \times 0,606 = 0,09$$

$$p(k=1 | n_1) = \frac{0,00438}{0,00438 + 0,09} = 0,0463$$

$$p(k=2 | n_1) = \frac{0,09}{0,00438 + 0,09} = 0,9537$$

$$\underline{n_2}: p(k | n_2) = \frac{p(n_2 | k) p(k)}{p(n_2)}$$

$$K=1: p(n_2 | k=1) \sim N \left(\mu_1 = \begin{bmatrix} 2,223 \\ -0,496 \end{bmatrix}, \Sigma_1 = \begin{bmatrix} 1,182 & -0,849 \\ -0,849 & 0,714 \end{bmatrix} \right) = \frac{1}{2\pi\sqrt{|\Sigma_1|}} \cdot e^{-1/2 (n_2 - \mu_1)^T \Sigma_1^{-1} (n_2 - \mu_1)}$$

$$|\Sigma_1| = \det \begin{bmatrix} 1,182 & -0,849 \\ 0,849 & 0,714 \end{bmatrix} = 0,123$$

$$\Sigma_1^{-1} = \frac{1}{0,123} \begin{bmatrix} 0,714 & 0,849 \\ 0,849 & 1,182 \end{bmatrix} = \begin{bmatrix} 5,805 & 6,886 \\ 6,886 & 9,618 \end{bmatrix}$$

$$n_2 - \mu_1 = \begin{bmatrix} 0 \\ 2 \end{bmatrix} - \begin{bmatrix} 2,223 \\ -0,496 \end{bmatrix} = \begin{bmatrix} -2,223 \\ 2,496 \end{bmatrix}$$

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① n_1

$$\frac{1}{2\pi\sqrt{0,123}} \cdot e^{-1/2 \cdot [-2,223 \quad 2,496] \cdot \begin{bmatrix} 9,586 & 6,886 \\ 6,886 & 5,792 \end{bmatrix} \begin{bmatrix} -2,223 \\ 2,496 \end{bmatrix}} = \frac{1}{2\pi\sqrt{0,123}} \cdot e^{-1/2 \cdot 7,045} \approx$$

$$\approx 0,01338$$

$$\text{Posterior}_1 = p(n_2 | k=1) \times \text{Prior}_1 = 0,01338 \times 0,394 = 0,005278$$

$$k=2 : p(n_2 | k=2) \sim N \left(\mu_2 = \begin{bmatrix} 0,754 \\ 0,873 \end{bmatrix}, \Sigma_2 = \begin{bmatrix} 0,947 & -1,0387 \\ -1,0387 & 1,364 \end{bmatrix} \right) = \frac{1}{2\pi\sqrt{|\Sigma_2|}} \cdot e^{-1/2 (n_2 - \mu_2)^T \cdot \Sigma_2^{-1} \cdot (n_2 - \mu_2)}$$

$$|\Sigma_2| = \det \begin{bmatrix} 0,947 & -1,0387 \\ -1,0387 & 1,364 \end{bmatrix} = 0,213$$

$$\Sigma_2^{-1} = \frac{1}{0,213} \cdot \begin{bmatrix} 1,364 & 1,0387 \\ 1,0387 & 0,947 \end{bmatrix} = \begin{bmatrix} 6,406 & 4,876 \\ 4,876 & 4,448 \end{bmatrix}$$

$$n_2 - \mu_2 = \begin{bmatrix} 0 \\ 2 \end{bmatrix} - \begin{bmatrix} 0,754 \\ 0,873 \end{bmatrix} = \begin{bmatrix} -0,754 \\ 1,127 \end{bmatrix}$$

$$= \frac{1}{2\pi\sqrt{0,213}} \cdot e^{-1/2 [-0,754 \quad 1,127] \cdot \begin{bmatrix} 6,406 & 4,876 \\ 4,876 & 4,448 \end{bmatrix} \begin{bmatrix} -0,754 \\ 1,127 \end{bmatrix}} = \frac{1}{2\pi\sqrt{0,213}} \cdot e^{-1/2 \cdot 1} \approx 0,209$$

$$\text{Posterior}_2 = p(n_2 | k=2) \times \text{Prior}_2 = 0,209 \times 0,606 = 0,1267$$

$$p(k=1 | n_2) = \frac{0,005278}{0,005278 + 0,1267} = 0,04$$

$$p(k=2 | n_2) = \frac{0,1267}{0,005278 + 0,1267} = 0,96$$

$$n_3 : p(k | n_3) = \frac{p(n_3 | k) \cdot p(k)}{p(n_3)}$$

$$k=1 : p(n_3 | k=1) \sim N \left(\mu_1 = \begin{bmatrix} 2,223 \\ -0,496 \end{bmatrix}, \Sigma_1 = \begin{bmatrix} 1,182 & -0,849 \\ -0,849 & 0,714 \end{bmatrix} \right) =$$

$$= \frac{1}{2\pi\sqrt{|\Sigma_1|}} \cdot e^{-1/2 (n_3 - \mu_1)^T \cdot \Sigma_1^{-1} \cdot (n_3 - \mu_1)}$$

$$|\Sigma_1| = \det \begin{bmatrix} 1,182 & -0,849 \\ -0,849 & 0,714 \end{bmatrix} = 0,123$$

$$\Sigma_1^{-1} = \frac{1}{0,123} \begin{bmatrix} 0,714 & 0,849 \\ 0,849 & 1,182 \end{bmatrix} = \begin{bmatrix} 5,804 & 6,886 \\ 6,886 & 9,617 \end{bmatrix}$$

$$n_3 - \mu_1 = \begin{bmatrix} 3 \\ -1 \end{bmatrix} - \begin{bmatrix} 2,223 \\ -0,496 \end{bmatrix} = \begin{bmatrix} 0,7767 \\ -0,5045 \end{bmatrix}$$

$$\frac{1}{2\pi\sqrt{0,123}} \cdot e^{-1/2 \cdot [0,7767 \quad -0,5045] \cdot \begin{bmatrix} 5,804 & 6,886 \\ 6,886 & 9,617 \end{bmatrix} \begin{bmatrix} 0,7767 \\ -0,5045 \end{bmatrix}} = \frac{1}{2\pi\sqrt{0,123}} \cdot e^{-1/2 \cdot 1,86} \approx 0,1788$$

$$\text{Posterior}_3 = p(n_3 | k=1) \times \text{Prior}_3 = 0,1788 \times 0,394 = 0,0705$$

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① cont.

$$K=2: p(n_3 | K=2) \sim N \left(\mu_2 = \begin{bmatrix} 0,754 \\ 0,873 \end{bmatrix}, \Sigma_2 = \begin{bmatrix} 0,947 & -1,0387 \\ -1,0387 & 1,361 \end{bmatrix} \right)$$

$$= \frac{1}{2\pi \sqrt{|\Sigma_2|}} \cdot e^{-1/2 \cdot (n_3 - \mu_2)^T \cdot \Sigma_2^{-1} \cdot (n_3 - \mu_2)} = \frac{1}{2\pi \sqrt{0,213}} \cdot e^{-1/2 \begin{bmatrix} 2,246 & 1,873 \end{bmatrix} \begin{bmatrix} 6,406 & 4,876 \\ 4,876 & 4,448 \end{bmatrix} \begin{bmatrix} 2,246 \\ -1,873 \end{bmatrix}}$$

$$\cdot |\Sigma_2| = \det \begin{bmatrix} 0,947 & -1,0387 \\ -1,0387 & 1,361 \end{bmatrix} = 0,213$$

$$\cdot \Sigma_2^{-1} = \frac{1}{0,213} \cdot \begin{bmatrix} 1,361 & 1,0387 \\ 1,0387 & 0,947 \end{bmatrix} = \begin{bmatrix} 6,406 & 4,876 \\ 4,876 & 4,448 \end{bmatrix}$$

$$\cdot n_3 - \mu_2 = \begin{bmatrix} 3 \\ -1 \end{bmatrix} - \begin{bmatrix} 0,754 \\ 0,873 \end{bmatrix} = \begin{bmatrix} 2,246 \\ -1,873 \end{bmatrix}$$

$$= \frac{1}{2\pi \sqrt{0,213}} \cdot e^{-1/2 \cdot 6,1889} \approx 0,011$$

$$\text{Posterior}_2 = p(n_3 | K=2) \times \text{Prior}_2 = 0,011 \times 0,00668 = 0,000668$$

$$\cdot \frac{P(K=1 | n_3) = 0,0705}{0,0705 + 0,000668} = 0,913$$

$$\cdot \frac{P(K=2 | n_3) = 0,000668}{0,0705 + 0,000668} = 0,087$$

M-STEP

$C=1$

$$\sum_{i=1}^N p(C | n_i) = 0,0463 + 0,04 + 0,913 = 1$$

$$\mu_1 = \frac{0,0463 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0,04 \begin{bmatrix} 0 \\ 2 \end{bmatrix} + 0,913 \begin{bmatrix} 3 \\ -1 \end{bmatrix}}{1} = \begin{bmatrix} 2,787 \\ -0,833 \end{bmatrix}$$

$$\sum_1^{(1,1)} = \frac{0,0463(1-2,787)^2 + 0,04(0-2,787)^2 + 0,913(3-2,787)^2}{1} \approx 0,1$$

$$\sum_1^{(2,1)} = \sum_1^{(1,2)} = \frac{0,0463(1-2,787)(0+0,833) + 0,04(0-2,787)(2+0,833) + 0,913(3-2,787)(-1+0,833)}{1} \approx -0,742$$

$$\sum_1^{(2,2)} = \frac{0,0463(0+0,833)^2 + 0,04(2+0,833)^2 + 0,913(-1+0,833)^2}{1} \approx 0,3786$$

Posterior₁

$$\Sigma_1 = \begin{bmatrix} 0,1 & -0,742 \\ -0,742 & 0,3786 \end{bmatrix}$$

$$\pi_1 = \frac{1}{3} \approx 0,333$$

① cont.

$$\sum_{i=1}^N P(L|n_i) = 0,9537 + 0,46 + 0,087 = 2$$

$$\mu_2 = \frac{0,9537 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0,46 \begin{bmatrix} 0 \\ 2 \end{bmatrix} + 0,087 \begin{bmatrix} 3 \\ -1 \end{bmatrix}}{2} = \begin{bmatrix} 0,6067 \\ 0,8734 \end{bmatrix}$$

$$\sum_2^{(1,1)} = \frac{0,9537(1-0,6067)^2 + 0,46(0-0,6067)^2 + 0,087(3-0,6067)^2}{2} \approx 0,5$$

$$\sum_2^{(1,1)} = \sum_2^{(1,2)} = \frac{0,9537(1-0,6067)(0-0,8734) + 0,46(0-0,6067)(2-0,8734) + 0,087(3-0,6067)(-1-0,8734)}{2} \approx -0,686$$

$$\sum_2^{(2,2)} = \frac{0,9537(0-0,8734)^2 + 0,46(2-0,8734)^2 + 0,087(-1-0,8734)^2}{2} \approx 1,125$$

Portanto

$$\Sigma_2 = \begin{bmatrix} 0,5 & -0,686 \\ -0,686 & 1,125 \end{bmatrix} \quad \pi_2 = \frac{2}{3} \approx 0,667$$

Para concluir, eis os valores atualizados da segunda epoch:

$$N_1 = 1$$

$$N_2 = 2$$

$$\mu_1 = \begin{bmatrix} 1,787 \\ -0,833 \end{bmatrix}$$

$$\mu_2 = \begin{bmatrix} 0,6067 \\ 0,8734 \end{bmatrix}$$

$$\Sigma_1 = \begin{bmatrix} 0,5 & -0,742 \\ -0,742 & 0,3786 \end{bmatrix}$$

$$\Sigma_2 = \begin{bmatrix} 0,5 & -0,686 \\ -0,686 & 1,125 \end{bmatrix}$$

$$\pi_1 = 0,333$$

$$\pi_2 = 0,667$$

② al Under MAP assumption, we must assign each observation to the highest posterior probability:

$$\bullet p(k=1|n_1) = 0,0463 \quad \bullet p(k=1|n_2) = 0,04 \quad \bullet p(k=1|n_3) = \underline{0,913}$$

$$\bullet p(k=2|n_1) = \underline{0,9537} \quad \bullet p(k=2|n_2) = \underline{0,96} \quad \bullet p(k=2|n_3) = 0,087$$

Therefore, n_3 is assigned to cluster 2, and n_1 and n_2 to cluster 1.

$$\text{clusters} = \{C_2 = \{n_1, n_2\}, C_1 = \{n_3\}\}$$

Pen-and-Paper

② b) Silhouette score for cluster C_2 :

$$S(C_2) = \frac{S(n_2) + S(n_1)}{2}$$

$$\bullet S(n_2) = \frac{b(n_2)}{a(n_2)} - 1 = \frac{\|n_2 - n_3\|_2}{\|n_2 - n_1\|_2} - 1 = \frac{\sqrt{18}}{\sqrt{5}} - 1 \approx 0,897$$

$$\left| \begin{array}{l} \bullet n_2 - n_3 = [0, 2] - [3, -1] = [-3, 3] \\ \|n_2 - n_3\|_2 = \sqrt{(-3)^2 + 3^2} = \sqrt{9+9} = \sqrt{18} \\ \bullet n_2 - n_1 = [0, 2] - [1, 0] = [-1, 2] \\ \|n_2 - n_1\|_2 = \sqrt{(-1)^2 + 2^2} = \sqrt{5} \end{array} \right.$$

$$\bullet S(n_1) = 1 - \frac{a(n_1)}{b(n_1)} = 1 - \frac{\|n_1 - n_2\|_2}{\|n_1 - n_3\|_2} = 1 - \frac{\sqrt{5}}{\sqrt{5}} = 0$$

$$\left| \begin{array}{l} \bullet n_1 - n_2 = [1, 0] - [0, 2] = [1, -2] \\ \|n_1 - n_2\|_2 = \sqrt{1^2 + (-2)^2} = \sqrt{5} \\ \bullet n_1 - n_3 = [1, 0] - [3, -1] = [-2, 1] \\ \|n_1 - n_3\|_2 = \sqrt{(-2)^2 + 1^2} = \sqrt{5} \end{array} \right.$$

$$\bullet S(C_2) = \frac{S(n_2) + S(n_1)}{2} = \frac{0,897 + 0}{2} \approx 0,449$$

Silhouette of largest cluster $S(C_2) = 0,449$.